PROBLEM STATEMENT BOOKLET

HACK & HIRE

Kickstart your career in **Sustainability**

This booklet covers the four companies setting problem statements for this round. Each section starts with a short introduction to the company, followed by its problem statements.

- **BlueNexus** — Water treatment
- **Bizsu** — Air conditioning efficiency
- **beez-fm** — Building decarbonisation
- **SystemEarth** — Supply chain compliance

ORGANISED BY Projet

SUPPORTED BY ATLNUS

BlueNexus Technologies

ABOUT THE COMPANY

AI-operated water treatment

BlueNexus Technologies is a water technology company based in Singapore. It builds AI systems that operate water treatment plants, so they need fewer staff on site, break down less often and cost less to run. Its systems work with the controls a plant already has.

AquaX Robot

An AI system that runs the whole plant. It uses camera, sound and infrared sensors to adjust treatment and flag equipment problems early.

AquaX Hub

A small on-site device that lets older plants and standalone equipment run with the same AI control.

i-WaterHub

A modular treatment plant built in a factory and operated by AquaX Robot.

PROBLEM STATEMENTS

Improving Water Treatment Efficiency Through Operational Data Analytics
Data Analytics / AI/ML / Engineering

Measuring and Communicating the Sustainability Impact of Autonomous Water Systems
ESG / Data Analytics

BLUENEXUS

Problem Statements

Role fit: Data Analytics / AI/ML / Engineering

Improving Water Treatment Efficiency Through Operational Data Analytics

BACKGROUND

Water treatment plants generate large amounts of operational data, including flow, pressure, energy use, chemical dosing, equipment status and water quality. However, identifying inefficient or abnormal operation can still depend heavily on manual monitoring and operator experience.

PROBLEM STATEMENT

How can BlueNexus use operational data to identify inefficient or abnormal plant operation and provide early insights that help reduce energy, chemical and water consumption while maintaining treatment performance?

Role fit: ESG / Data Analytics

Measuring and Communicating the Sustainability Impact of Autonomous Water Systems

BACKGROUND

BlueNexus develops AI-operated water treatment systems designed to improve water reuse, reduce energy consumption and minimise chemical usage. As clients and regulators increasingly require sustainability reporting, there is a need for a clear and credible way to quantify and communicate these benefits.

PROBLEM STATEMENT

How can BlueNexus measure and present the sustainability impact of its autonomous water systems in a way that is credible and useful for clients' ESG reporting?

ABOUT THE COMPANY

Air conditioners that use less electricity

Bizsu is a climate technology company started in Malaysia and now working in Malaysia and Singapore. Its main product is a patented natural ceramic net that helps air conditioners cool a room using less electricity.

Installation

The net is placed above the air conditioner's return filter. The unit does not need to be modified or switched off.

How it works

It reduces moisture in the air passing through, so the room cools down faster.

Result

The air conditioner runs for less time, which lowers electricity use and bills.

Patented product

Used in offices, hotels, warehouses and data centres

Malaysia and Singapore

PROBLEM STATEMENTS

Build a Scalable Lead-Generation and Sales Engine
Business Development / AI Engineering

Build Brand Awareness Through High-Impact Content
Marketing & Comms

BIZSU

Problem Statements

Role fit: Business Development / AI Engineering

Build a Scalable Lead-Generation and Sales Engine

BACKGROUND

Bizsu makes a patented ceramic filter for air conditioners that helps them cool faster and use less electricity, without any changes to the unit. Most of its customers so far are businesses and large buildings. Bizsu now wants to reach homeowners, who need a simple and convincing reason to try the product.

PROBLEM STATEMENT

How might we develop a scalable strategy to generate qualified leads and drive sales among homeowners?

Teams may explore cold emailing a sizeable list of relevant people per team, targeting a strong positive response rate, while using AI tools and free resources. They may also choose to create a conversion-focused website, with or without AI, and apply effective copywriting techniques that are direct, concise, and storytelling-driven.

Role fit: Marketing & Comms

Build Brand Awareness Through High-Impact Content

BACKGROUND

Bizsu's product sits inside the air conditioner where nobody sees it, and its benefit only shows up later on the electricity bill. This makes it hard to explain in a quick post or ad. Bizsu wants content that explains the product clearly and introduces the team behind it.

PROBLEM STATEMENT

How might we increase brand awareness and motivate potential customers to take action through engaging content?

Teams may choose to create and host a professional webinar targeting a solid audience, as well as produce creative videos about the business, people, and products. They are encouraged to experiment with innovative, fun, and unconventional formats. Brice, the CEO, may be involved in selected videos where beneficial.

ABOUT THE COMPANY

making decarbonization obvious.

beez-fm runs a marketplace in Asia Pacific for cutting energy use and emissions in buildings, built on its RizaOne platform. Building owners pay nothing upfront for upgrades to systems such as air conditioning, lighting and controls. Green investors fund the upgrades, and they are paid back from the energy savings, which are checked every month.

“You buy the outcome, not the equipment.”

Connect
Collects live data from the building’s management system

›

Measure
Runs an automated energy audit of the building

›

Propose
Recommends technologies based on expected savings

›

Deliver
Installs the upgrade, paid back from measured savings

PROBLEM STATEMENT

Visualising Technology Fit to Help Building Owners Choose the Right Efficiency Solution
Data Visualisation

Problem Statements

Role fit: Data Visualisation

Visualising Technology Fit to Help Building Owners Choose the Right Efficiency Solution

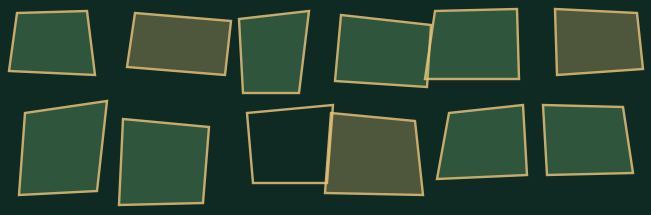
BACKGROUND

beez-fm's RizaOne platform matches each connected building to the right efficiency technology using that building's own energy data. Before a building owner agrees to a zero-CAPEX pilot, they need to see why a specific technology was recommended for their building rather than another, and trust that recommendation.

PROBLEM STATEMENT

How can beez-fm visualise a building's energy profile and technology match so that owners can clearly see and trust why a specific recommendation fits their building?

SystemEarth



ABOUT THE COMPANY

Sustainable sourcing, sorted.

SystemEarth is a Singapore-based software platform that helps companies check that their suppliers meet sustainability and due diligence rules. Its customers are manufacturers and traders with international supply chains, especially in natural rubber and timber.

One platform. Every framework.
Collect once, validate once, report everywhere.

Plot verification
Checks and fixes the farm plot boundaries that suppliers submit

Deforestation risk
Compares each plot with forest cover and forest loss maps

Regulatory mapping
Checks plots against national land use and boundary data

Supplier data
An online portal for collecting supplier information and following up on issues

EUDR · CSDDD · Human rights due diligence

PROBLEM STATEMENT

Building Multi-Layer, Multi-Year Geospatial Data
Geospatial / Data Engineering

SYSTEMEARTH

Problem Statements

Role fit: Geospatial / Data Engineering

Building Multi-Layer, Multi-Year Geospatial Data

BACKGROUND

SystemEarth's evaluation of geospatial risk for farm plots is a composite of multiple geospatial layers, such as tree cover loss, natural forest baselines and restricted use areas, among others. Yet, a compliance decision rarely needs the entire raw value of data, tending to require segments of pixel information: presence of a forest at a particular plot of land at the regulatory cut-off date, degradation over time, overlapping geometry, social, environmental and governance risks etc, creating a challenge of handling vast amounts of information in a performant and accurate manner.

PROBLEM STATEMENT

Present on a specific use case and demonstrate how SystemEarth can represent multiple geospatial layers across time and space in a compact and query-efficient form, while preserving the information required to reproduce plot-level compliance decisions?